Bihar Engineering University, Patna

B.Tech 5th Semester Examination, 2024

Course: B.Tech
Code: 105501

Subject: Artificial Intelligence

Time: 03 Hours
Full Marks: 70

Instructions:-

(i) The marks are indicated in the right-hand margin.
(ii) There are NINE questions in this paper.
(iii) Attempt FIVE questions in all.
(iv) Question No. 1 is compulsory.

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Q.1 ***Choose the correct option / answer the following (Any seven question only):*** **[2 x 7 = 14]**

(a) Which of the following best defines the Turing Test?
- (i) A method to calculate machine efficiency
- (ii) A test for machine learning algorithms
- (iii) A test to determine if a machine can exhibit human-like intelligence
- (iv) A benchmark for robotic speed

(b) *The A** algorithm guarantees optimality if the heuristic used is:
- (i) Arbitrary
- (ii) Inconsistent
- (iii) Admissible and consistent
- (iv) Random

(c) Which of these search strategies is *not* complete in infinite-depth spaces?
- (i) Breadth-First Search
- (ii) Depth-First Search
- (iii) DFID
- (iv) A* Search

(d) In an AND/OR graph, solving an AND node means:
- (i) Solving one child node is enough
- (ii) All child nodes must be solved
- (iii) The node is ignored
- (iv) Any node can be skipped

(e) Minimax is used in game playing to:
- (i) Maximize randomness
- (ii) Minimize evaluation time
- (iii) Make optimal moves assuming the opponent plays optimally
- (iv) Store all possible states

(f) Alpha-beta pruning improves Minimax by:
- (i) Increasing the depth
- (ii) Ignoring non-optimal branches
- (iii) Doubling the search time
- (iv) Reducing the score of nodes

(g) Propositional logic cannot express which of the following?
- (i) True/false statements
- (ii) Compound statements
- (iii) Logical connectives
- (iv) Variables and functions

(h) Bayesian networks are especially useful in situations with:
- (i) Complete knowledge and logic
- (ii) No uncertainty
- (iii) Deterministic processes
- (iv) Probabilistic and uncertain information

(i) Which of the following is most likely used in deep learning?
- (i) Decision Trees
- (ii) Shallow neural networks
- (iii) Multi-layered neural networks
- (iv) K-Means algorithm

(j) Which of the following is a key step in a genetic algorithm?
- (i) Regression
- (ii) Cross-validation
- (iii) Crossover
- (iv) Tokenization

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